Group 2: Project Specifications

Project Title: NLP Based Smart Mock Interview Assistant: A System to assess Technical and Behavioural interview response.

Project topic	AI-powered mock interview simulator for technology-related roles
Research area	Natural Language Processing, Machine Learning, Software Development
Team GitHub Repository	https://github.com/ekanth3128/SMART-MOCK-INTERVIEW.git
Target users	Students, graduates and early career applicants preparing for technical interviews

1. Project Topic

This project will develop an AI-powered mock interview assistant for technology-related roles such as Data Analyst, Data Scientist, Software Engineer and AI/ML Engineer. The system will present technical and behavioural interview questions, evaluate the typed answers of the user, and give personalized feedback and scores of improvement.

The project aims to solve a problem students and graduates/early career candidates face: interview preparation guides do offer questions and model answers but fail to provide consistent, objective, and role-specific feedback to the student's answer.

2. Research Question

Can neural NLP models accurately evaluate technical and behavioural interview responses and provide personalised feedback in a mock interview simulator for technology-related roles?

3. Problem Statement

Candidates preparing for technology interviews may not know whether their answers are correct, complete, relevant or professionally expressed. Existing resources usually support passive revision rather than interactive practice. This creates a gap between reading example answers and receiving realistic feedback on personal interview performance.

This project will build a mock interview assistant that compares user responses with reference answers and scoring criteria, then generates structured feedback on correctness, relevance, clarity, completeness and interview readiness.

4. Data Sources

Source	Content	Planned use
Kaggle Python Interview Questions 2025	Python-focused technical interview questions and answers.	Technical question bank and reference answers
SQL Interview Preparation GitHub Repository	SQL interview questions, preparation notes, and database-related question content.	SQL technical question bank, role/topic tagging for Data Analyst/Data Scientist roles, and reference answer support.
Data Science, ML, DL and NLP Interview Questions GitHub Repository	Interview questions covering data science, machine learning, deep learning, and NLP topics.	Core technical question bank for Data Scientist and AI/ML Engineer roles, topic classification, and reference answer development
Non-technical Interview Questions and Answers GitHub Repository	Behavioural, HR, and general non-technical interview questions and answers.	Behavioural question bank, soft-skill response examples, rubric development, and feedback generation for non-technical answers

Table 1. Data Sources

Note: the datasets are mainly interview question and answer datasets and may not include labelled candidate answer quality scores. Therefore, the project will define a scoring framework using semantic similarity, keyword/concept coverage, relevance, clarity and completeness.

5. Project Objectives

1. Collect, clean and structure interview question data from at least two datasets or sources.
2. Create a question bank with role categories, technical/behavioural labels, topic tags and reference answers.
3. Develop a response scoring framework for correctness, relevance, clarity, completeness, concept coverage and readiness.
4. Implement and compare at least three models: a neural network, a hybrid model and a pre-trained/transfer learning model. Week 5 work begins the pre-trained model route using Sentence-BERT.
5. Critically analyze the results of a model using appropriate measures like F1 score, accuracy, cosine similarity, correlation to rubric scores and mean absolute error.
6. Tune hyperparameters, or adjust thresholds, refine features in the model or select from a different model to optimize the model.
7. Build a user facing app that allows role selection, interview practice, answer submission, scoring and feedback generation.

6. Modelling Approach

Model type	Example approach	Purpose
Neural network	LSTM, BiLSTM, CNN-text or feed-forward neural model using embeddings	Learn response quality patterns from engineered labels or rubric-based scores
Hybrid model	Semantic similarity + keyword/concept coverage + clarity rules + ML classifier/regressor	Provide explainable scoring and feedback criteria
Pre-trained / transfer learning	Sentence-BERT embeddings using sentence-transformers/all-MiniLM-L6-v2	Measure semantic relevance between a user's answer and the reference answer, especially where labelled candidate-answer quality data is limited.

Table 2. Modelling Approach

7. Goals

1. The aim is to provide a working mock interview assistant that will assess written responses and give beneficial and individualized feedback. The goal is not to replace human interviewers, but to provide a realistic coaching aid that provides students with clear, consistent, and enabling instructions.
2. A functional prototype of the Web App with selection of roles, delivery of questions, submission of answers, automated scoring and feedback.
3. A tested scoring framework covering technical correctness, relevance, clarity, completeness and readiness.
4. A comparison of at least three models with a justified best-performing approach.
5. Clear final documentation explaining the dataset pipeline, model results, limitations and future improvements.

8. KPIs / Key Performance Indicators

Area	KPI	Success measure
Model performance	Three model comparison completed	Neural, hybrid and pre-trained/transfer learning models implemented and evaluated; Week 5 pre-trained Sentence-BERT component started.
Model performance	Scoring quality	Best model shows acceptable accuracy/F1, similarity score, correlation or MAE against validation rubric. Week 5 pretrained output is semantic similarity.

Model performance	Technical answer evaluation	System identifies missing key concepts and relevant technical coverage
Model performance	Behavioural answer evaluation	System evaluates clarity, relevance, structure and completeness of behavioural answers
App success	Functional interview flow	User can select role, answer questions, complete a session and view results
App success	Feedback usefulness	Feedback is personalised, readable and gives specific improvement actions
App success	Usability and response time	Interface is clear and feedback returns within a reasonable time
Project success	Dataset requirement met	At least two sources are cleaned, documented and used in the pipeline
Project success	Optimisation completed	At least one tuning/refinement stage is recorded and compared
Project success	Final deliverables completed	Working app, model evaluation, GitHub documentation and final report are submitted

Table 3. KPIs / Key Performance Indicators

9. Expected Final Output and Scope

The final product will be a web application that simulates an interview for the user with an NLP based scoring system. The user will learn the interview questions, provide written responses, get computerized marks and receive personalized feedback.

The project scope is limited to written responses rather than live speech or audio. The system will not make hiring decisions. It will act as an interview preparation tool, with transparent scoring criteria and explainable feedback to reduce the risk of over-reliance on a single automated score.